



Faculty of Engineering & Applied Science

Modelling and Simulation Assignment 1
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Problem Statement

The objective of this model is to create a store checkout system where given a single cashier as well as 20, 40, or 60 customers for each scenario, key metrics are established for analytical purposes.

Assumptions

Table 1: Assumptions Summary

Assumption	Explanation
Random seed	An arbitrary seed value of 11 was used for reproducibility of results.
Interarrival time	Assume equiprobability (no peak hour assumption for simplicity). To execute this a uniform distribution was used as a generator with units as the number of minutes found on [1,10]. Intuition is, 10 minutes is slower periods and closer to 1 minute is a busier period.
Checkout time	A discrete distribution with values [2, 3, 4, 5, 6, 7, 8] minutes and corresponding [0.04, 0.13, 0.30, 0.25, 0.15, 0.09, 0.04] probabilities. This was crafted in such a way as to follow a bell curve shape. On average 4.3 minutes is generated which is sensible as it is neither too high nor too low. In addition, no values are negative as time>0.
First-In-First-Out (First-Come-First-Serve)	Sequential ordering of arrival i.e. first person goes in and gets priority and then leaves first as well.
No stopping in between.	Once a customer begins, they go until they finish. There are no leave early scenarios at play. No psychology is mimicked at all.
Fixed customer scenarios	The three cases are exactly 20, 40, and 60 customers showing up rather than using those metrics as an upper boundary and generating random scenarios of that number showing up. It is purely static.
Number of cashiers is dynamically built but defaults to 1	Users can input the number of cashiers with the base value being normally 1 as that is the requirement but other integers may be entered to see different outcomes and to better understand the overall system. If

	anything other than an integer is entered then it'll default to the required 1.
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Probability Tables

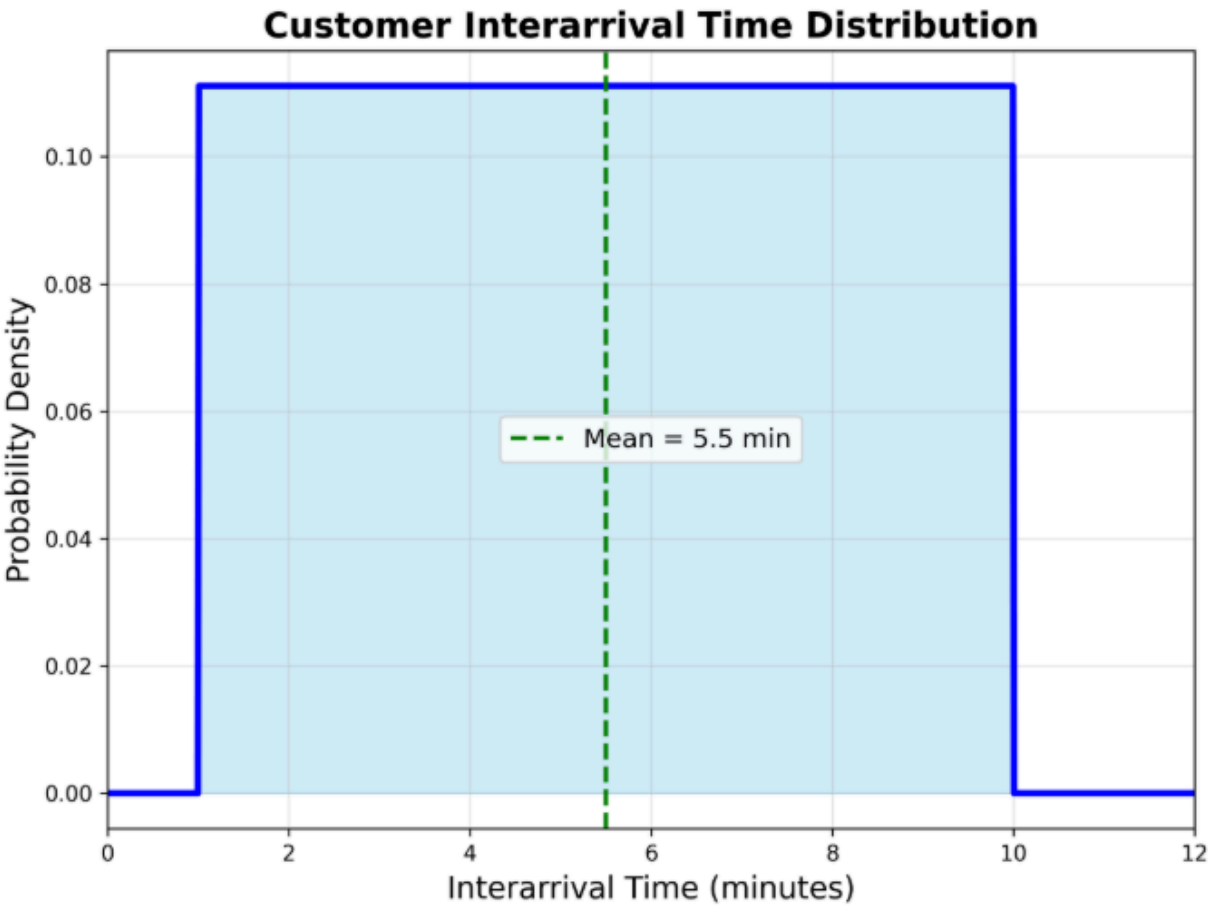


Figure 1: Uniform distribution bound on a min of 1 and a max of 10 minutes as parameters.

A uniform distribution was used and generated with the lower bound being 1 minute and the most upper bound being 10 minutes. This is a continuous distribution for the assumption in the model.

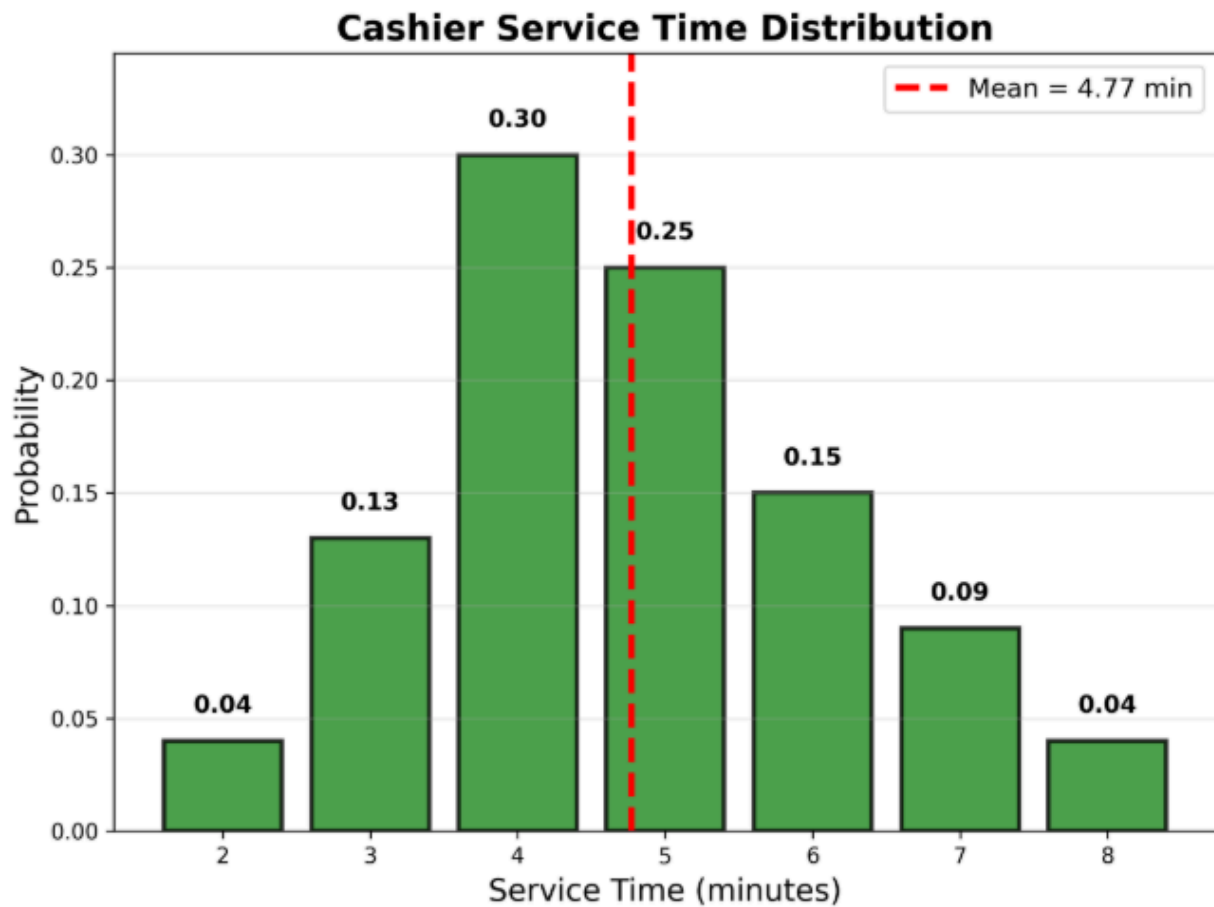


Figure 2: Discrete service time distribution graph.

For the discrete service time distribution, a discrete distribution was utilized which maps the number of integer minute scenarios from 2 to 8 with associated probabilities that are somewhat reminiscent of a bell curve to improve robustness as less scenarios will be the lower or most upper bound.

Table 2: Discrete probability distribution representing cashier service time.

Probability	0.04	0.13	0.3	0.25	0.15	0.09	0.04
Random variable (Minutes)	2	3	4	5	6	7	8

Results

```
Please input the number of cashiers simulated: 1
Case 20 metrics: num_customers=20.0000, avg_waiting_time=2.0455, avg_time_in_system=6.3955, probability_of_waiting=0.4500, cashier_utilization=0.7202, idle_time=33.7964, idle_time_percentage=0.2798
Case 40 metrics: num_customers=40.0000, avg_waiting_time=2.6789, avg_time_in_system=7.3039, probability_of_waiting=0.6000, cashier_utilization=0.8404, idle_time=35.1404, idle_time_percentage=0.1596
Case 60 metrics: num_customers=60.0000, avg_waiting_time=6.0792, avg_time_in_system=10.9626, probability_of_waiting=0.8167, cashier_utilization=0.9141, idle_time=27.5326, idle_time_percentage=0.0859

Comparative analysis (1cashier(s) for three different scenarios including 20, 40, and 60 customers injected into the system)
Case 20: The average wait time is 2.0455, The average time in system is 6.3955, The probability of waiting is 0.4500, The cashier utilization is 0.7202, The idle time percentage is 0.2798
Case 40: The average wait time is 2.6789, The average time in system is 7.3039, The probability of waiting is 0.6000, The cashier utilization is 0.8404, The idle time percentage is 0.1596
Case 60: The average wait time is 6.0792, The average time in system is 10.9626, The probability of waiting is 0.8167, The cashier utilization is 0.9141, The idle time percentage is 0.0859

Congestion Metric:
20 to 40 customers: Waiting increased by 0.6334 minutes, Time in system increased by 0.9084 minutes
40 to 60 customers: Waiting increased by 3.4003 minutes, Time in system increased by 3.6587 minutes
20 to 60 customers: Waiting increased by 4.0337 minutes, Time in system increased by 4.5671 minutes
```

Figure 3: Results for having a single cashier.

Above shows all the results given that there is only one cashier working. For each of the three static scenarios (i.e. 20, 40, 60) the numbers of customers, average waiting time, average time in, probability of waiting, cashier utilization, idle time, and idle time percentage are all given. For the comparative analysis results, the average wait time, average time in the system, the probability of waiting, cashier utilization and idle time percentage are outputted. Finally, for the congestion metrics, comparing the step changes between the 20, 40 and 60 customer scenarios is done for which the waiting differential, and time in system differential are found.

```
Please input the number of cashiers simulated: 2
Case 20 metrics: num_customers=20.0000, avg_waiting_time=0.0438, avg_time_in_system=4.3938, probability_of_waiting=0.1000, cashier_utilization=0.7202, idle_time=33.7964, idle_time_percentage=0.2798
Case 40 metrics: num_customers=40.0000, avg_waiting_time=0.0277, avg_time_in_system=4.6527, probability_of_waiting=0.0250, cashier_utilization=0.8426, idle_time=34.5684, idle_time_percentage=0.1574
Case 60 metrics: num_customers=60.0000, avg_waiting_time=0.0942, avg_time_in_system=4.9776, probability_of_waiting=0.1000, cashier_utilization=0.9257, idle_time=23.5326, idle_time_percentage=0.0743

Comparative analysis (2cashier(s) for three different scenarios including 20, 40, and 60 customers injected into the system)
Case 20: The average wait time is 0.0438, The average time in system is 4.3938, The probability of waiting is 0.1000, The cashier utilization is 0.7202, The idle time percentage is 0.2798
Case 40: The average wait time is 0.0277, The average time in system is 4.6527, The probability of waiting is 0.0250, The cashier utilization is 0.8426, The idle time percentage is 0.1574
Case 60: The average wait time is 0.0942, The average time in system is 4.9776, The probability of waiting is 0.1000, The cashier utilization is 0.9257, The idle time percentage is 0.0743

Congestion Metric:
20 to 40 customers: Waiting increased by -0.0161 minutes, Time in system increased by 0.2589 minutes
40 to 60 customers: Waiting increased by 0.0665 minutes, Time in system increased by 0.3249 minutes
20 to 60 customers: Waiting increased by 0.0505 minutes, Time in system increased by 0.5838 minutes
```

Figure 4: Results for having two cashiers for comparison.

```
Please input the number of cashiers simulated: 3
Case 20 metrics: num_customers=20.0000, avg_waiting_time=0.0000, avg_time_in_system=4.3500, probability_of_waiting=0.0000, cashier_utilization=0.7202, idle_time=33.7964, idle_time_percentage=0.2798
Case 40 metrics: num_customers=40.0000, avg_waiting_time=0.0000, avg_time_in_system=4.6250, probability_of_waiting=0.0000, cashier_utilization=0.8426, idle_time=34.5684, idle_time_percentage=0.1574
Case 60 metrics: num_customers=60.0000, avg_waiting_time=0.0000, avg_time_in_system=4.8833, probability_of_waiting=0.0000, cashier_utilization=0.9265, idle_time=23.2603, idle_time_percentage=0.0735

Comparative analysis (3cashier(s) for three different scenarios including 20, 40, and 60 customers injected into the system)
Case 20: The average wait time is 0.0000, The average time in system is 4.3500, The probability of waiting is 0.0000, The cashier utilization is 0.7202, The idle time percentage is 0.2798
Case 40: The average wait time is 0.0000, The average time in system is 4.6250, The probability of waiting is 0.0000, The cashier utilization is 0.8426, The idle time percentage is 0.1574
Case 60: The average wait time is 0.0000, The average time in system is 4.8833, The probability of waiting is 0.0000, The cashier utilization is 0.9265, The idle time percentage is 0.0735

Congestion Metric:
20 to 40 customers: Waiting increased by 0.0000 minutes, Time in system increased by 0.2750 minutes
40 to 60 customers: Waiting increased by 0.0000 minutes, Time in system increased by 0.2583 minutes
20 to 60 customers: Waiting increased by 0.0000 minutes, Time in system increased by 0.5333 minutes
```

Figure 5: Results for having three cashiers for comparison.

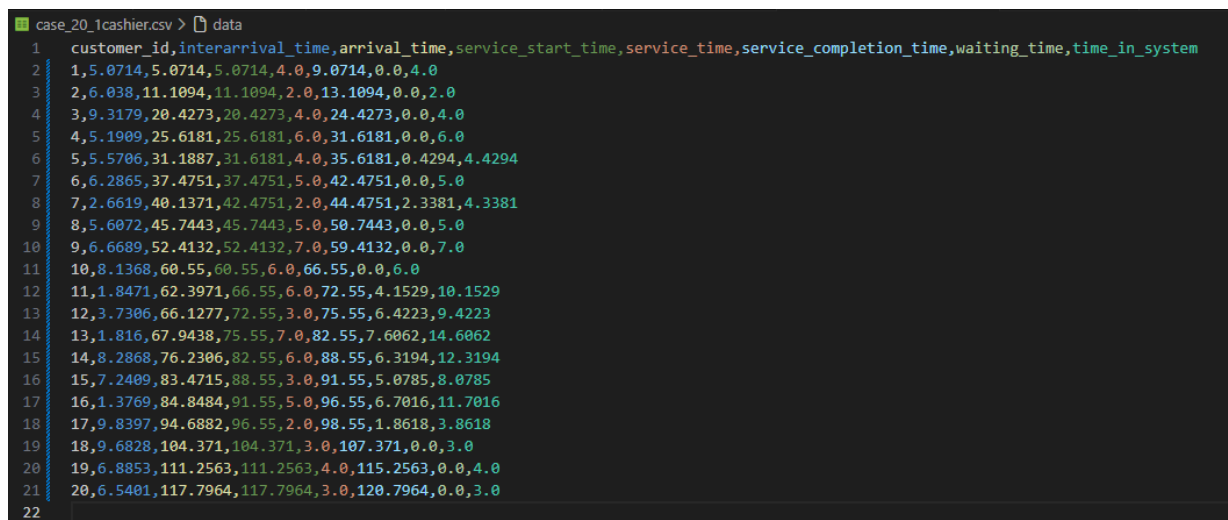
Figures 4 and 5 show the same outputs but for two and three cashiers respectively. Comparing all results based on this model's assumptions, when there are either 1 or 2 cashiers, the probabilities of waiting are non-zero as well as the average wait time indicating that there is some excess traffic in a given moment. But when there are three cashiers, the probability of waiting is zero as well as the average wait time hence indicating that having three cashiers is sufficient to not have any holdups and to have an efficient system for all of

the cases. This makes sense mathematically, to run through the reason why this is to look at the single cashier scenario for 20 customers. The cashier utilization is 0.7202 and the idle capacity is 0.2798. In addition, the average service wait is 2.4055 minutes and the probability of waiting is 45%. Similarly comparing these with the other customer scenarios, it can be seen that there is nonlinear growth when looking at similar metrics for the other scenarios. There is deterioration in regard to having adequate cashier staff as the number of customers trends up, leading to an insufficient result for the case where there is only 1 cashier.

Looking at the congestion results, especially honing in on the main analysis which is having one cashier, the average wait time jumps significantly i.e. a couple minutes each increment when comparing the 20 customer scenario, 40 customer scenario and 60 customer scenario.

Cashier utilization was found by taking the sum of service times and the simulation end time and taking a ratio. Moving from 20 to 40 to 60 customers, there is naturally an increase.

Table of Results Sample:



customer_id	interarrival_time	arrival_time	service_start_time	service_time	service_completion_time	waiting_time	time_in_system
1	5.0714	5.0714	5.0714	4.0	9.0714	0.0	4.0
2	6.038	11.1094	11.1094	2.0	13.1094	0.0	2.0
3	9.3179	20.4273	20.4273	4.0	24.4273	0.0	4.0
4	5.1909	25.6181	25.6181	6.0	31.6181	0.0	6.0
5	5.5706	31.1887	31.6181	4.0	35.6181	0.4294	4.4294
6	6.2865	37.4751	37.4751	5.0	42.4751	0.0	5.0
7	2.6619	40.1371	42.4751	2.0	44.4751	2.3381	4.3381
8	5.6072	45.7443	45.7443	5.0	50.7443	0.0	5.0
9	6.6689	52.4132	52.4132	7.0	59.4132	0.0	7.0
10	8.1368	60.55	60.55	6.0	66.55	0.0	6.0
11	1.8471	62.3971	66.55	6.0	72.55	4.1529	10.1529
12	3.7306	66.1277	72.55	3.0	75.55	6.4223	9.4223
13	1.816	67.9438	75.55	7.0	82.55	7.6062	14.6062
14	8.2868	76.2306	82.55	6.0	88.55	6.3194	12.3194
15	7.2409	83.4715	88.55	3.0	91.55	5.0785	8.0785
16	1.3769	84.8484	91.55	5.0	96.55	6.7016	11.7016
17	9.8397	94.6882	96.55	2.0	98.55	1.8618	3.8618
18	9.6828	104.371	104.371	3.0	107.371	0.0	3.0
19	6.8853	111.2563	111.2563	4.0	115.2563	0.0	4.0
20	6.5401	117.7964	117.7964	3.0	120.7964	0.0	3.0

Figure 6: Table for case 20 with one cashier.

See appendix B for other table scenarios (i.e. 40 and 60 customers). The customer ID, interarrival time, arrival time, service start time, service time, service completion time, waiting time, and the time in the system.

Recommendation To Improve Model

After reflecting on the results and noting reality, there is an array of improvements that can be done and are listed as follows:

- Have variance in the timing of day. For example if peak time is say from 1PM-2PM then this will carry a larger probability weighting for more customers to come.
- The number of customers are currently static i.e. 20, 40, 60, one way to improve is to have those as maximums and randomly generate under to apply more randomness.

- Having psychological thresholds is vital to make it more realistic as people have randomness and varying personalities. For example, have a threshold that says, if a customer shows up, and waits for 10 minutes, they can get impatient and leave after that allotted time.
- During testing of my results there was a scenario when I used a seed of 50 where the waiting time differential comparing 20 and 40 was negative indicating that there were less customers in scenario 1, but it took longer than the 40 customers scenario in regards to waiting time.

References

[1] "Basic concepts," Basic Concepts - SimPy 4.1.2.dev8+g81c7218 documentation, https://simpy.readthedocs.io/en/latest/simpy_intro/basic_concepts.html (accessed Feb. 13, 2026).

Appendices

Appendix A: Miscellaneous Planning

Assignment 1

① Initialize System

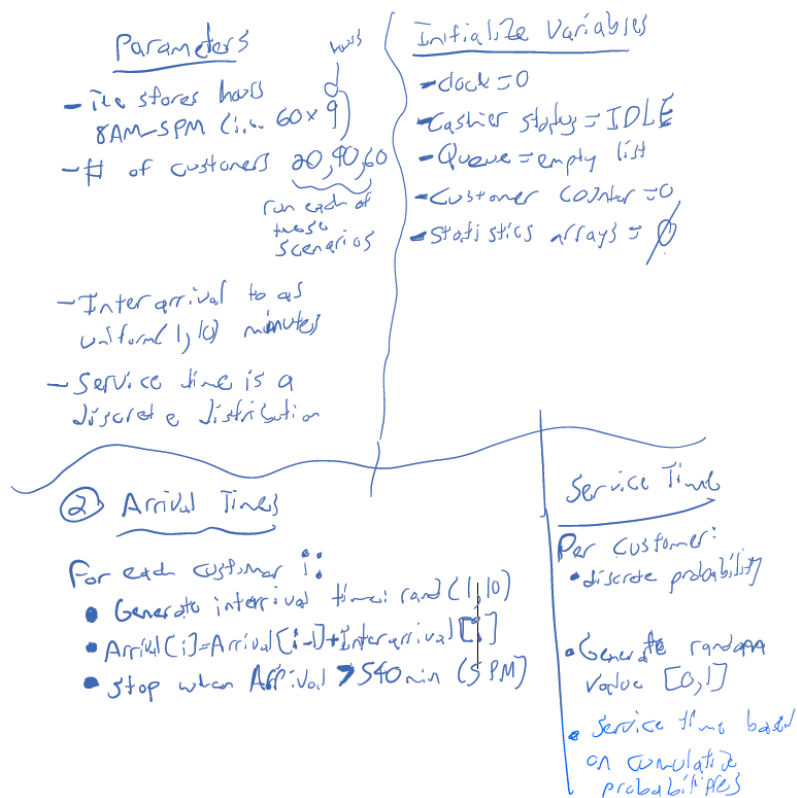


Figure 7: Rough brainstorming.

Appendix B: Extra Results

```

case_40_1cashier.csv > data
1 customer_id,interarrival_time,arrival_time,service_start_time,service_time,service_completion_time,waiting_time,time_in_system
2 1,2.4174,2.4174,2.4174,6.0,8.4174,0.0,6.0
3 2,1.135,3.5525,8.4174,6.0,14.4174,4.865,10.865
4 3,5.7554,9.3079,14.4174,4.0,18.4174,5.1096,9.1096
5 4,1.536,10.8438,18.4174,4.0,22.4174,7.5736,11.5736
6 5,2.7119,13.5557,22.4174,3.0,25.4174,8.8617,11.8617
7 6,3.1775,16.7332,25.4174,3.0,28.4174,8.6842,11.6842
8 7,1.2707,18.0039,28.4174,5.0,33.4174,10.4135,15.4135
9 8,5.1754,23.1794,33.4174,5.0,38.4174,10.2381,15.2381
10 9,4.9648,28.1441,38.4174,3.0,41.4174,10.2733,13.2733
11 10,8.5818,36.726,41.4174,5.0,46.4174,4.6915,9.6915
12 11,5.6721,42.3981,46.4174,4.0,50.4174,4.0193,8.0193
13 12,6.7626,49.1607,50.4174,6.0,56.4174,1.2567,7.2567
14 13,5.498,54.6587,56.4174,5.0,61.4174,1.7588,6.7588
15 14,6.962,61.6207,61.6207,5.0,66.6207,0.0,5.0
16 15,5.116,66.7367,66.7367,5.0,71.7367,0.0,5.0
17 16,3.5035,70.2402,71.7367,7.0,78.7367,1.4965,8.4965
18 17,9.9789,80.2191,80.2191,8.0,88.2191,0.0,8.0
19 18,9.9612,90.1803,90.1803,4.0,94.1803,0.0,4.0
20 19,8.5619,98.7422,98.7422,4.0,102.7422,0.0,4.0
21 20,7.3703,106.1125,106.1125,6.0,112.1125,0.0,6.0
22 21,3.8375,109.95,112.1125,3.0,115.1125,2.1625,5.1625
23 22,3.067,113.017,115.1125,4.0,119.1125,2.0955,6.0955
24 23,3.6014,116.6184,119.1125,3.0,122.1125,2.4942,5.4942
25 24,1.632,118.2504,122.1125,4.0,126.1125,3.8621,7.8621
26 25,7.8966,126.147,126.147,3.0,129.147,0.0,3.0
27 26,4.6036,130.7506,130.7506,4.0,134.7506,0.0,4.0
28 27,8.6193,139.3698,139.3698,6.0,145.3698,0.0,6.0
29 28,4.4786,143.8484,145.3698,7.0,152.3698,1.5214,8.5214
30 29,9.6224,153.4708,153.4708,6.0,159.4708,0.0,6.0
31 30,8.6258,162.0966,162.0966,5.0,167.0966,0.0,5.0
32 31,1.0049,163.1015,167.0966,5.0,172.0966,3.9951,8.9951
33 32,2.8875,165.989,172.0966,4.0,176.0966,6.1076,10.1076
34 33,9.1924,175.1814,176.0966,3.0,179.0966,0.9152,3.9152
35 34,5.2299,180.4113,180.4113,4.0,184.4113,0.0,4.0
36 35,9.8232,190.2345,190.2345,6.0,196.2345,0.0,6.0
37 36,4.5768,194.8114,196.2345,3.0,199.2345,1.4232,4.4232
38 37,1.6573,196.4687,199.2345,2.0,201.2345,2.7658,4.7658
39 38,6.6651,203.1338,203.1338,6.0,209.1338,0.0,6.0
40 39,8.0066,211.1404,211.1404,4.0,215.1404,0.0,4.0
41 40,3.428,214.5684,215.1404,5.0,220.1404,0.572,5.572
42

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Figure 8: Results for 40 customers and 1 cashier.

```

customer_id,interarrival_time,arrival_time,service_start_time,service_time,service_completion_time,waiting_time,time_in_system
1,1.7843,1.7843,1.7843,6.0,7.7843,0.0,6.0
2,3.9933,5.7776,7.7843,7.0,14.7843,2.0067,9.0067
3,9.6767,15.4543,15.4543,4.0,19.4543,0.0,4.0
4,7.8224,23.2766,23.2766,4.0,27.2766,0.0,4.0
5,2.0619,25.3385,27.2766,6.0,33.2766,1.9381,7.9381
6,3.2175,28.556,33.2766,6.0,39.2766,4.7206,10.7206
7,1.9094,30.4655,39.2766,5.0,44.2766,8.8112,13.8112
8,1.539,32.0045,44.2766,6.0,50.2766,12.2721,18.2721
9,8.1732,40.1777,50.2766,5.0,55.2766,10.0989,15.0989
10,2.5991,42.7768,55.2766,7.0,62.2766,12.4998,19.4998
11,6.0337,48.8104,62.2766,4.0,66.2766,13.4662,17.4662
12,5.0268,53.8373,66.2766,6.0,72.2766,12.4393,18.4393
13,2.7162,56.5534,72.2766,7.0,79.2766,15.7232,22.7232

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14,7.587,64.1405,79.2766,4.0,83.2766,15.1361,19.1361
15,2.1787,66.3192,83.2766,5.0,88.2766,16.9574,21.9574
16,6.7934,73.1126,88.2766,4.0,92.2766,15.164,19.164
17,2.0486,75.1612,92.2766,4.0,96.2766,17.1154,21.1154
18,4.7868,79.948,96.2766,3.0,99.2766,16.3286,19.3286
19,2.9158,82.8638,99.2766,2.0,101.2766,16.4128,18.4128
20,3.4282,86.2919,101.2766,6.0,107.2766,14.9847,20.9847
21,9.7384,96.0303,107.2766,4.0,111.2766,11.2463,15.2463
22,8.2307,104.261,111.2766,5.0,116.2766,7.0156,12.0156
23,3.7373,107.9983,116.2766,6.0,122.2766,8.2783,14.2783
24,8.9638,116.9621,122.2766,4.0,126.2766,5.3145,9.3145
25,2.8964,119.8585,126.2766,4.0,130.2766,6.4181,10.4181
26,4.5485,124.407,130.2766,3.0,133.2766,5.8697,8.8697
27,8.6894,133.0963,133.2766,6.0,139.2766,0.1803,6.1803
28,6.7765,139.8729,139.8729,6.0,145.8729,0.0,6.0
29,1.903,141.7759,145.8729,6.0,151.8729,4.097,10.097
30,9.9037,151.6796,151.8729,7.0,158.8729,0.1933,7.1933
31,2.9192,154.5988,158.8729,5.0,163.8729,4.2741,9.2741
32,3.3245,157.9233,163.8729,5.0,168.8729,5.9496,10.9496
33,7.9542,165.8775,168.8729,4.0,172.8729,2.9954,6.9954
34,3.9606,169.8381,172.8729,6.0,178.8729,3.0348,9.0348
35,3.6669,173.505,178.8729,4.0,182.8729,5.3679,9.3679
36,1.6606,175.1656,182.8729,4.0,186.8729,7.7073,11.7073
37,1.8111,176.9766,186.8729,4.0,190.8729,9.8962,13.8962
38,6.2446,183.2213,190.8729,5.0,195.8729,7.6516,12.6516
39,3.1871,186.4084,195.8729,5.0,200.8729,9.4645,14.4645
40,6.4116,192.8199,200.8729,5.0,205.8729,8.0529,13.0529
41,4.3453,197.1653,205.8729,4.0,209.8729,8.7076,12.7076
42,5.0789,202.2441,209.8729,4.0,213.8729,7.6287,11.6287
43,9.6322,211.8763,213.8729,6.0,219.8729,1.9965,7.9965
44,5.3535,217.2299,219.8729,4.0,223.8729,2.643,6.643
45,6.1711,223.401,223.8729,4.0,227.8729,0.4719,4.4719
46,8.7987,232.1997,232.1997,3.0,235.1997,0.0,3.0
47,2.6454,234.8452,235.1997,4.0,239.1997,0.3546,4.3546
48,2.3872,237.2324,239.1997,3.0,242.1997,1.9673,4.9673
49,9.1758,246.4082,246.4082,4.0,250.4082,0.0,4.0
50,8.3602,254.7684,254.7684,5.0,259.7684,0.0,5.0
51,3.2455,258.0139,259.7684,8.0,267.7684,1.7545,9.7545
52,2.7082,260.7221,267.7684,5.0,272.7684,7.0463,12.0463
53,7.6548,268.3769,272.7684,4.0,276.7684,4.3915,8.3915

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54,9.4636,277.8406,277.8406,5.0,282.8406,0.0,5.0
55,2.7693,280.6099,282.8406,5.0,287.8406,2.2307,7.2307
56,9.5512,290.1611,290.1611,6.0,296.1611,0.0,6.0
57,8.9397,299.1008,299.1008,5.0,304.1008,0.0,5.0
58,6.4318,305.5326,305.5326,7.0,312.5326,0.0,7.0
59,4.7931,310.3258,312.5326,4.0,316.5326,2.2069,6.2069
60,1.9346,312.2603,316.5326,4.0,320.5326,4.2723,8.2723
```

Figure 9: Results for 60 customers and 1 cashier.